

Tilak Mudgal

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(Open to Relocate)

Experienced **Business Analyst** with 3+ years of experience in transforming complex data into actionable insights to support strategic decision-making and operational efficiency. Proficient in **SQL**, **Python**, and data visualization tools such as **Power BI** and **Tableau**. Skilled in collaborating with stakeholders to bridge business needs with data-driven solutions, enhancing process efficiency and driving business growth.

Actively seeking opportunities.

PROFESSIONAL EXPERIENCE

QUBER INC

Remote, US

Business Analyst

May 2023 - Dec 2024

Project 1: Income Model Analysis

Developed an income predictive model to accurately estimate income levels using data derived from bureau and bank statements, employing advanced machine learning techniques and feature engineering

- Extracted, cleaned, and processed large-scale financial data from **MongoDB** and **MySQL** ensuring data accuracy and consistency using **Python (Pandas, NumPy)** for effective analysis and reporting
- Optimized **SQL** queries using **stored procedures** and **window functions** to improve data retrieval efficiency, reduce execution time, and streamline reporting processes for large-scale financial datasets
- Conducted **exploratory data analysis (EDA)** using Python (**Matplotlib**, **Seaborn**) to identify trends, outliers, and correlations, leveraging descriptive statistics and hypothesis testing for actionable insights
- Applied **statistical techniques** such as **correlation analysis**, **VIF**, and **feature scaling** to optimize feature selection from 200+ variables, improving data quality and model inputs
- Designed **interactive dashboards** in **Tableau** and automated reporting workflows to visualize financial trends, enabling data-driven decisions and enhancing business intelligence
- Collaborated with cross-functional teams to analyze model performance using **Python (Scikit-learn)**, applying regression analysis, KFold cross-validation, and achieving an **R-squared** of 0.93

Project 2: Dry Eye and Punctal Plug Market Research

Conduct predictive analytics of the market share of eye care products in the United States, to determine the viability of selling niche eye care products and acquisition of assets.

- Collected, processed, and managed large datasets using **AWS S3** ensuring data integrity and improving query performance for business analysis
- Automated data extraction and preprocessing workflows with **Python (Pandas, NumPy)** and Apache Airflow, reducing manual effort by 20% and enhancing operational efficiency
- Conducted **exploratory data analysis (EDA)** using **Amazon Redshift** to identify trends, resolve data inconsistencies, and improve data quality
- Analyzed market trends and consumer behavior using **AWS QuickSight**, identifying a 15% growth opportunity, supporting business expansion strategies
- Developed interactive dashboards and reports to provide actionable insights and revenue projections, enabling data-driven decision-making for executive stakeholders

MYSHUBHLIFE

Bengaluru, India

Data Analyst

Oct 2020 - June 2022

Project 1: Loan Disbursement and Company Performance Dashboard

Developed a real-time dashboard to address the need to track loan performance metrics (applications, approvals, rejections, and defaults) and forecast future trends. The solution improved early risk detection by 15% and enabled data-driven decisions through predictive modeling and interactive visualizations

- Developed a comprehensive **Power BI** dashboard utilizing **DAX** functions and Excel to track overall loan disbursement performance for over a **million** customers, including total disbursed amount of **\$100M+**, applications received, approved, rejected, and default rates.
- Created predictive models using **SQL** and **Python** to forecast future loan performance, integrating key **KPIs** like **P&L**, cash flow, and default risk, resulting in a 15% improvement in early risk detection.
- Contributed to the successful **migration** from **Power BI** to **AWS QuickSight**, optimizing report performance, reducing operational costs, and ensuring seamless integration with SQL-based data sources for real-time analytics.
- Collaborated with senior leadership and finance teams to present loan performance insights, fostering alignment between data findings and business objectives through effective **storytelling** and **stakeholder engagement**.
- Automated reporting workflows using **Power BI, SQL, and Excel** (Pivot Tables, VBA, Power Query), reducing manual reporting efforts by 30% and improving report turnaround time.

Project 2: Enhancing Customer Retention

*Conducted an **A/B testing** to evaluate the effectiveness of personalized in-app notifications versus standard email campaigns for re-engaging inactive users, using SQL and Python for data analysis and experiment design.*

- Conducted **A/B testing** to compare personalized in-app notifications with email campaigns, to analyze user engagement and measure performance across key metrics.
- Achieved a **25%** increase in **re-engagement rates**, an **15%** boost in **loan applications**, and a rise in premium feature adoption, contributing to an improvement in overall retention.
- Applied **Z-test** for large-scale analysis, ensuring statistical rigor and validating the superior effectiveness of in-app notifications with a **p-value < 0.05**, supporting data-driven decision-making.
- Built interactive **Power BI** dashboards to provide real-time visibility into critical performance metrics, including **session durations**, and **retention rates**, enabling stakeholders to track experiment outcomes and optimize future strategies.

Project 3: Customer Segmentation for Credit Policy Enhancement

The Credit Risk Assessment and Segmentation Tool was developed to enhance credit policy effectiveness and optimize marketing strategies by accurately identifying distinct customer segments based on financial behavior analysis. The tool utilized advanced decision tree modeling and hierarchical clustering techniques to assess creditworthiness and tailor policies accordingly

- Employed an **empirical model** combining decision tree modeling and **hierarchical clustering** to perform customer segmentation analysis. This rigorous approach Determined 4 distinct customer segments (A, B, C, and D) based on relevant variables, significantly enhancing the precision of credit policies
- Leveraged bureau scores and extracted features from 6-month bank statements, such as transaction amounts and account balances, to assess customer's financial capabilities.
- Leveraged hierarchical clustering to categorize customers into segments with varying creditworthiness and risk levels. Applied **decision tree** modeling to pinpoint the most influential factors in credit risk assessment, enabling data-driven decisions.
- Established empirical decision-making rules based on the analysis, leading to a 30% increase in targeted marketing effectiveness and a 10% reduction in default rates. The integration of these rules with bank statement analysis optimized resource allocation and Optimized overall credit policy outcomes.
- Coordinated with cross-functional teams to integrate the tool into existing credit evaluation systems, enhancing utility and adoption across the organization. This led to more informed investment decisions and better resource management

PROJECTS

AI-Powered University Chatbot Development

- Built an AI-driven university chatbot leveraging the **BERT model** and AMD ROCm GPU, currently trained on 2,000 dialogues and 10,000 question-answer pairs.
- Integrated the AMD Instinct™ MI210 accelerator card and the **Large Language Model (LLM)** architecture to optimize AI performance, expediting chatbot responses, and enhancing the overall user experience.

AI Voice Assistant Hub: Multi-Role Conversational AI Platform

- Developed a voice-based AI assistant using **OpenAI's GPT-3.5** and **Whisper APIs**, supporting multiple specialized roles (e.g., Restaurant, Banking, Healthcare) with text-to-speech responses and conversation history tracking.
- Implemented a user-friendly **Streamlit** interface enabling seamless toggling between continuous and manual recording modes, facilitating real-time interactions and personalized user experiences.

SKILLS

- **Databases:** MySQL, Transact SQL, PostgreSQL, MongoDB
- **Programming Languages and Tools:** Python, R, SQL, SAS, Matlab, DAX, Tableau, Power BI, Advanced Excel (Pivot Tables, Lookups, VBA)
- **Business Skills:** Stakeholder Management, A/B Testing, Data Storytelling, Business Process Optimization, Market Research, Jira, Financial Analysis, Agile Methodologies, Data Governance, Requirements Gathering, Business Process Optimization
- **Cloud:** AWS (S3, Glue, QuickSight), Azure (Data Lake Gen 2, Data Factory, Azure Databricks)
- **Machine Learning:** Linear & Logistic Regression, scikit-learn, Random Forests, Decision Trees, Neural, Natural Language Processing (NLP), XGBoost, Statistical Modeling, Hypothesis Testing
- **Framework & Libraries:** NumPy, Pandas, Scikit-Learn, Matplotlib, LangChain, PyTorch, Streamlit

EDUCATION

UNIVERSITY OF MASSACHUSETTS DARTMOUTH
MS in Data Science (GPA: 3.83/4)

Massachusetts, US
Aug 2022 - Aug 2024

SIDDAGANGA INSTITUTE OF TECHNOLOGY
B.E Information Science and Engineering

Karnataka, India
Aug 2015 - Oct 2020